Pre-work · Blood Vessels and Blood

Blood Vessels, Blood Vessel Disorders and Fetal Circulation, and Blood. Read all three Part A chapters first. Most of this is histology, so work it with a slide or an image beside you.

Module 3 · 3 of 5 | Bring this to class

How to work this pre-work

The order matters more than the hours. Context first, content second. Most students who struggle did every piece of this, in the wrong sequence.

1. **Read** Read the Part A chapter first. This packet does not repeat it. It puts it to work.
2. **Organize** Panel 1. Turn the reading into mini visuals. Not copied sentences, your structure.
3. **Draw** Panel 2. Brain dump from memory, then draw, then correct in a second colour.
4. **Apply** Panel 3. Notes closed for the first pass. Open them only to check.
5. **Retrieve** Later in the week, not the same night. Cover everything and redraw your mini visuals from memory. What you cannot redraw is what you do not yet know.

Why this order. Reading before you organize gives you context. Organizing before you draw forces you to decide what connects to what. Drawing before you apply is where you find out what you actually know. If you skip to the application questions, you will answer them with the notes open and learn almost nothing.

PANEL 1 OF 3

Organize it

Turn the vessel and blood chapters into mini visuals. Eight chunks. Two grids do the histology and one chain does the fetal detour.

FIVE SHAPES, THAT IS THE WHOLE TOOLKIT

Information has a shape. Pick the wrong one and the visual is noise. Pick the right one and you can redraw it from memory in twenty seconds. Ladder for an ordered series. Chain for a sequence, with arrows. Split for a pair you keep confusing. Hub for one thing with parts hanging off it. Grid for two variables crossed.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk

PANEL 2 OF 3

Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

PANEL 3 OF 3

Apply it

First pass with everything closed. Then open the notes in a second colour and mark what you missed.

WORKED EXAMPLE

A slide shows a vessel with a wide, irregular lumen, an incomplete basement membrane, and very large gaps between the endothelial cells. Whole blood cells are passing through the wall. Name the vessel and one organ it is found in.

Answer. A sinusoid. Liver, spleen, or red bone marrow.

Reasoning. Only one vessel in the body has an incomplete basement membrane, and that alone settles it. Work back from what is crossing: a continuous capillary keeps most substances out, a fenestrated capillary passes fluid and small solutes through pores, but whole cells need gaps far larger than either. So the tissue must be one where cells legitimately move between blood and tissue, which is exactly what the liver, spleen, and marrow do. Structure is matched to the job, and the leakiest wall goes where the largest thing has to cross.

takes longer than three minutes you are copying instead of organizing.

1 **GRID**

The three tunics

Tunic down the side. Position, tissue, and one distinguishing note across the top. Add a fourth column: is it present in a capillary, yes or no.

2 **GRID**

The five vessel types

Vessel down the side, in the order blood meets them. Direction, wall, and role across the top. This grid is the frame for the whole chapter.

3 **SPLIT**

Artery against vein, on a slide

One split. Wall thickness, lumen shape, tunica media, tunica externa, and valves. This is the pair you will be asked to tell apart, so make the giveaway column obvious.

4 **GRID**

The three capillary types

Capillary down the side, wall and where found across the top. Add a fourth column: what can cross it. Three rows, and the third column is the reasoning column.

5 **CHAIN**

Fetal circulation, placenta to placenta

One long chain with arrows. Umbilical vein, ductus venosus, right atrium, then the two ways past the lungs, and back out the umbilical arteries. Mark each shunt where it appears.

DRAWING 1

An artery and a vein, side by side in cross section

BRAIN DUMP FIRST

One minute. All three tunics in each, and every difference between them.

Draw the two vessels beside each other, cut across. Label all three tunics in each, and draw the internal and external elastic laminae where they belong. Make the artery wall thick with a round lumen and the vein wall thin with a wide, collapsed lumen. Add a valve to the vein and vasa vasorum to the externa of both.

In your notebook, head the page **M3-1 An artery and a vein, side by side in cross section** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

Someone shows you an unlabelled slide and you name the vessel in under five seconds, and can say which feature decided it.

Set A · Name it

1. The only tunic a capillary has is the ______________________.
2. The small vessels supplying the wall of a large vessel are the ______________________.
3. The ring of smooth muscle controlling entry to a capillary is the ______________________.
4. The adult remnant of the ductus arteriosus is the ______________________.
5. Name the three main plasma proteins and one job of each.

Set B · Predict it

1. A deep vein thrombosis forms in the calf and breaks free. Trace the exact route it takes and name where it lodges.
2. The valves of the leg veins fail. Predict what you would see and explain it from venous pressure.
3. The ductus arteriosus fails to close after birth. Predict the abnormal connection this leaves.
4. An abnormal hemoglobin makes a red cell rigid. Predict where in the circulation it causes trouble and why.

Set C · Tell it apart

1. Explain why blood is drawn from a vein rather than an artery, using three structural differences.

6 **GRID**

Fetal shunt to adult remnant

Fetal structure down the side, what it does and what it becomes across the top. Five rows. Circle the one that does not become a cord.

7 **HUB**

Blood, composition

One spun tube at the hub. Spoke to plasma with its three proteins, the buffy coat, and the erythrocyte layer. Write the percentage beside each layer.

8 **GRID**

The five leukocytes

Leukocyte down the side. Group, nucleus, granules, and defense role across the top. Order the rows most to least numerous, because that order is itself an answer.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

DRAWING 2

The fetal circulation, with the three shunts

BRAIN DUMP FIRST

Two minutes. Every fetal vessel and where each detour goes.

Draw a simple fetus with the heart, liver, and lungs marked, and the placenta beside it. Draw the umbilical vein in, the ductus venosus past the liver, the foramen ovale between the atria, the ductus arteriosus between the pulmonary trunk and the aorta, and the two umbilical arteries out. Arrow the whole route. Then in a second colour write the adult remnant beside each shunt.

In your notebook, head the page **M3–2 The fetal circulation, with the three shunts** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can explain why each shunt exists by pointing at the organ it bypasses and saying what that organ is not yet doing.

2. Distinguish an aneurysm from an atherosclerotic plaque by what each does to the vessel wall.

3. Blood is called a connective tissue. Defend that classification using its matrix.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colours, and you can say out loud why each correction was a correction.

DRAWING 3

A blood smear

BRAIN DUMP FIRST

One minute. Every formed element, drawn as it looks under the microscope.

Draw a field of a blood smear. Fill most of it with biconcave erythrocytes, showing the pale centre. Add one of each leukocyte, drawn by its nucleus: multi-lobed neutrophil, bilobed eosinophil, granule-obscured basophil, round-nucleus lymphocyte, kidney-shaped monocyte. Scatter platelets between them. Label each and note its relative number.

In your notebook, head the page **M3–3 A blood smear** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can identify each cell from the nucleus alone, with the labels covered.

Mini visual sheet

One frame per chunk, shaped for what goes in it. Tall frames are ladders, wide strips are chains, squares are hubs. Draw straight onto this sheet. If a frame is too small for your handwriting, redraw that one in your notes and put the page number on the line.

1 The three tunics

GRID

2 The five vessel types

GRID

3 Artery against vein, on a slide

SPLIT

4 The three capillary types

GRID

5 Fetal circulation, placenta to placenta

CHAIN

6 Fetal shunt to adult remnant

GRID

7 Blood, composition

HUB

8 The five leukocytes

GRID